

SAT Math Mastery · Week 5B

Ratios, Rates, Units & Density

Time 150 minutes: 110 min solving + 40 min self-check and error log

Topics unit conversions (including squared and cubed units) · rates and rate expressions · density · ratios and proportions · scale drawings and scale factors · dimensional analysis

Name / Date _____

1 · SKILLS & IDEAS

- S1.** Convert units with a chain of conversion factors so that unwanted units cancel.
- S2.** Convert squared and cubed units: use the length factor squared for area and cubed for volume.
- S3.** Convert compound rates (distance per time, volume per time, anything per time²).
- S4.** Set up and solve proportions; split a total in a given ratio; compare unit rates.
- S5.** Scale drawings and maps: lengths scale by k , areas by k^2 , volumes by k^3 .
- S6.** Density: $\rho = \frac{m}{V}$; find mass, volume or edge length; combine materials by mass or by volume.
- S7.** Build symbolic rate expressions (“at this rate, how many...”) and test them with numbers.
- S8.** Hidden conditions: relative motion, combined work rates, mixtures, similar solids.

2 · REQUIRED FACTS

- A conversion factor equals 1 (for example $\frac{100 \text{ cm}}{1 \text{ m}}$); multiply so the unit you want to remove sits in the opposite position and cancels.
- If lengths convert by the factor c , then areas convert by c^2 and volumes by c^3 . Rates over time² use the time factor squared.
- 1 L = 1000 cm³, 1 m³ = 1000 L, 1 m³ = 10⁶ cm³, 1 h = 3600 s, 1 km/h = $\frac{1}{3.6}$ m/s.
- Scale 1 : k (or “1 unit represents k units”): actual length = $k \times$ drawn length; actual area = $k^2 \times$ drawn area.
- Densities do not average: density of a mixture is $\frac{\text{total mass}}{\text{total volume}}$.
- Combined work rates add: if A needs a hours and B needs b hours, together they do $\frac{1}{a} + \frac{1}{b}$ of the job per hour.

3 · REQUIRED FORMULAS

$$\begin{aligned} \text{rate} &= \frac{\text{quantity}}{\text{time}}, & d &= rt, & \rho &= \frac{m}{V}, & m &= \rho V, & V_{\text{cube}} &= s^3, & A_{\text{cube}} &= 6s^2 \\ \frac{\text{part}}{\text{whole}} &= \frac{a}{a+b} \text{ for ratio } a:b, & \frac{A_2}{A_1} &= k^2, & \frac{V_2}{V_1} &= k^3, & \frac{m_2}{m_1} &= k^3 \text{ (same material)} \\ V_{\text{cylinder}} &= \pi r^2 h, & \text{flow rate} &= \text{cross-section area} \times \text{speed}, & \rho_{\text{mix}} &= \frac{m_1 + m_2}{V_1 + V_2} \end{aligned}$$

4 · MINI THEORY SHEET

Unit chains. Write the starting quantity, then multiply by fractions that equal 1, each cancelling one unit.

Example: $72 \frac{\text{km}}{\text{h}} \times \frac{1000 \text{ m}}{1 \text{ km}} \times \frac{1 \text{ h}}{3600 \text{ s}} = 20 \frac{\text{m}}{\text{s}}$.

Squared and cubed units. 1 m = 100 cm $\Rightarrow$ 1 m² = 100² cm² = 10⁴ cm² and 1 m³ = 10⁶ cm³. Converting 0.6 km² to m² multiplies by 1000², not 1000.

Scale. Drawing-to-actual factor k for length gives k^2 for area and k^3 for volume or mass. To go between two maps, convert through the actual size, or multiply the two length factors and square.

Ratios and rates. A ratio $a : b$ splits a total T into $\frac{a}{a+b}T$ and $\frac{b}{a+b}T$. For “at this rate” questions, find the unit rate first, then scale.

Density. Mass = density $\times$ volume. To find an edge, find the volume first and then take the cube root. Mixtures: add masses and add volumes, then divide.

Symbolic rates. Write the rate with its units (boxes per minute), convert the time to match, multiply. Verify with simple numbers.

Speed shortcuts.

Situation	Shortcut
area / volume units	multiply by the length factor ² / length factor ³
km/h $\leftrightarrow$ m/s	divide by 3.6 / multiply by 3.6
rate over time ²	the time factor is squared: $\text{m/s}^2 \rightarrow \text{km/h}^2$ multiplies by $\frac{1}{1000} \cdot 3600^2$
scale factor from area	$k = \sqrt{\text{area ratio}}$; then mass or volume ratio is k^3
ratio $a : b$ of a total	fraction $\frac{a}{a+b}$ of the total; do not divide by b
catch-up problems	head start = (speed difference) $\times$ time
symbolic rate check	substitute easy numbers and test each option
mixtures	$\rho_{\text{mix}} = \frac{\text{total mass}}{\text{total volume}}$, never the average of densities

5 · GUIDED EXAMPLES

Example 1 — Cubed units

A tank holds 2.4 m^3 of water. How many liters is that? ($1 \text{ m} = 100 \text{ cm}$, $1 \text{ L} = 1000 \text{ cm}^3$)

$$2.4 \text{ m}^3 = 2.4 \times 100^3 \text{ cm}^3 = 2.4 \times 10^6 \text{ cm}^3. \text{ Then } \frac{2.4 \times 10^6}{1000} = 2400 \text{ L.}$$

Example 2 — Scale and area

On a map, 1 cm represents 5 km . A lake covers 6 cm^2 on the map. Find its actual area.

Lengths scale by 5, so areas scale by $5^2 = 25$: $6 \times 25 = 150 \text{ km}^2$ (not $6 \times 5 = 30$).

Example 3 — Density of a mixture

A 60 g piece of metal with density 12 g/cm^3 is melted with a 30 g piece with density 3 g/cm^3 . Find the density of the mixture (volumes add).

$$\text{Volumes: } \frac{60}{12} + \frac{30}{3} = 5 + 10 = 15 \text{ cm}^3. \text{ Mass: } 90 \text{ g. Density: } \frac{90}{15} = 6 \text{ g/cm}^3, \text{ not the average } 7.5.$$

6 · PRACTICE SET — 46 PROBLEMS · EASY → MEDIUM → HARD → SAT CHALLENGE

Each problem states its tool. Write your work in the space below. Multiple choice: one correct option. Student-produced response (SPR): enter an exact value (fraction or decimal).

1. EASY · CALCULATOR

A car travels at a constant speed and covers 270 miles in 4.5 hours. At this rate, how many miles will the car travel in 7 hours?

2. EASY · CALCULATOR

A board is 30 inches long. Given that 1 inch = 2.54 centimeters, what is the length of the board, in centimeters?

- (A) 32.54 (B) 76.2
(C) 762 (D) 11.81

3. EASY · CALCULATOR

A metal block has a volume of 12 cm^3 and a mass of 94.8 grams. What is the density of the metal, in grams per cubic centimeter?

- (A) 1137.6 (B) 0.127
(C) 7.9 (D) 82.8

4. EASY · NO CALCULATOR NEEDED

The table shows map distances and the corresponding actual distances for a map that has a constant scale. What actual distance, in kilometers, corresponds to a map distance of 7 centimeters?

Map distance (cm)	2	5	8
Actual distance (km)	30	75	120

5.

EASY · NO CALCULATOR NEEDED

A conveyor belt moves at a constant speed of 0.8 meters per second. At this speed, how many meters does the belt move in 1 hour?

(A) 2880

(B) $\frac{1}{4500}$

(C) 48

(D) 288

6.

MEDIUM · NO CALCULATOR NEEDED

A rectangular rug has an area of 2.5 square feet. Given that 1 foot = 12 inches, what is the area of the rug, in square inches?

(A) 4320

(B) $\frac{2.5}{144}$

(C) 30

(D) 360

7.

MEDIUM · NO CALCULATOR NEEDED

A storage tank has a volume of 3.5 cubic meters. How many liters of water does the tank hold when full? (1 meter = 100 centimeters and 1 liter = 1000 cubic centimeters.)

8.

MEDIUM · NO CALCULATOR NEEDED

A train travels at a constant speed of 72 kilometers per hour. What is the speed of the train, in meters per second?

- (A) 259.2 (B) 1.2
(C) 20 (D) 72

9.

MEDIUM · NO CALCULATOR NEEDED

On a scale drawing of a building, 0.75 inch represents 6 feet. A wall measures 4.5 inches on the drawing. What is the actual length of the wall, in feet?

- (A) 27 (B) 36
(C) $\frac{9}{16}$ (D) 6

10.

MEDIUM · CALCULATOR

Aluminum has a density of 2.7 grams per cubic centimeter. What is the mass, in grams, of a solid aluminum cube with edge length 5 centimeters?

- (A) 13.5 (B) 40.5
(C) 67.5 (D) 337.5

11.

MEDIUM · CALCULATOR

Flour and sugar are mixed in the ratio 5 to 3 by mass. A batch of the mixture has a total mass of 4.8 kilograms. How many kilograms of sugar are in the batch?

12.

MEDIUM · NO CALCULATOR NEEDED

A machine fills b boxes every m minutes. At this rate, which of the following expressions gives the number of boxes the machine fills in h hours?

(A) $\frac{60bh}{m}$

(B) $\frac{60hm}{b}$

(C) $\frac{bm}{60h}$

(D) $\frac{bh}{60m}$

13.

MEDIUM · NO CALCULATOR NEEDED

Pump A can fill a tank in 8 hours working alone. Pump B can fill the same tank in 24 hours working alone. How many hours will it take the two pumps, working together, to fill the tank?

14.

HARD · NO CALCULATOR NEEDED

Water leaks from a reservoir at a constant rate of 2.4 cubic meters per hour. At this rate, how many liters of water leak out per minute? (1 cubic meter = 1000 liters.)

- (A) 4 (B) 40
(C) 144 (D) 2400

15.

HARD · CALCULATOR

Solar panels, each covering 2 square meters, are installed so that they cover 40% of a solar farm. The farm has an area of 0.6 square kilometers. How many panels are installed? (1 kilometer = 1000 meters.)

16.

HARD · CALCULATOR

A solid lead cube has a mass of 2440.8 grams. The density of lead is 11.3 grams per cubic centimeter. What is the edge length of the cube, in centimeters?

- (A) 72 (B) 216
(C) 6 (D) 36

17.

HARD · CALCULATOR

A model of a bridge is built at a scale of 1 : 40. The total surface area of the model is 150 square centimeters. What is the surface area of the actual bridge, in square meters?

- (A) 24 (B) 2400
(C) 240,000 (D) 0.6

18.

HARD · CALCULATOR

Vehicle A travels at 54 kilometers per hour, vehicle B at 16 meters per second, and vehicle C at 0.93 kilometers per minute. Which of the following lists the vehicles in order from slowest to fastest?

(A) B, C, A

(B) A, B, C

(C) C, A, B

(D) A, C, B

19.

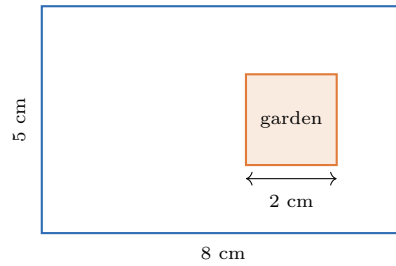
HARD · NO CALCULATOR NEEDED

A mixture is made of concentrate and water in the ratio 2 to 7 by volume, and 18 liters of it is prepared. How many liters of concentrate must be added to the 18 liters so that the ratio of concentrate to water becomes 1 to 2?

20.

HARD · NO CALCULATOR NEEDED

The scale drawing shows a rectangular plot of land that contains a square garden. On the drawing, 1 centimeter represents 3 meters. What is the actual area, in square meters, of the part of the plot that is *not* the garden?



(A) 36

(B) 108

(C) 324

(D) 972

21.

HARD · CALCULATOR

A car travels 32 miles per gallon of fuel. Use 1 mile = 1.609 kilometers and 1 gallon = 3.785 liters. To the nearest tenth, how many kilometers does the car travel per liter of fuel?

22.

HARD · NO CALCULATOR NEEDED

The edge length of a cube is tripled. Which of the following statements must be true?

- I. The surface area of the cube is multiplied by 9.
- II. The volume of the cube is multiplied by 27.
- III. The ratio of surface area to volume is multiplied by 3.

(A) I only

(B) I and II only

(C) II and III only

(D) I, II and III

23.

HARD · CALCULATOR

A ball's speed increases at a constant rate of 9.8 meters per second each second, that is, an acceleration of 9.8 m/s^2 . What is this acceleration in kilometers per hour squared (km/h^2)?

24.

SAT CHALLENGE · NO CALCULATOR NEEDED

Metal X has a density of 8.0 g/cm^3 and metal Y has a density of 3.0 g/cm^3 . An alloy is made by combining X and Y in the mass ratio 3 : 2 (X to Y). Assuming the volumes simply add, what is the density of the alloy, in g/cm^3 ?

(A) 5.0

(B) 5.5

(C) 6.0

(D) 4.8

25.

SAT CHALLENGE · NO CALCULATOR NEEDED

On Map P, 1 centimeter represents 2 kilometers. On Map Q, 1 centimeter represents 5 kilometers. On Map P, a park occupies 40% of a square region whose side measures 7.5 centimeters. What is the area of the park on Map Q, in square centimeters?

26.

SAT CHALLENGE · CALCULATOR

A freight train leaves a station traveling at a constant 60 kilometers per hour. 1.5 hours later, a second train leaves the same station on the same track, in the same direction, at a constant 25 meters per second. How many kilometers from the station will the second train be when it catches up to the first train?

27.

SAT CHALLENGE • CALCULATOR

Water flows through a cylindrical pipe with a circular cross section of radius 5 centimeters at a constant speed of 2 meters per second. To the nearest liter, how many liters of water pass through the pipe in one minute? (1 L = 1000 cm³.)

28.

SAT CHALLENGE · NO CALCULATOR NEEDED

Two solid statues are geometrically similar and made of the same uniform material. The smaller statue has a mass of 16 kilograms and a surface area of 0.8 square meters. The larger statue has a surface area of 5 square meters. What is the mass of the larger statue, in kilograms?

(A) 250

(B) 400

(C) 40

(D) 100

29.

SAT CHALLENGE · NO CALCULATOR NEEDED

A car travels m miles per gallon of fuel, and fuel costs d dollars per gallon. Given that 1 mile = 1.6 kilometers, which of the following expressions gives the fuel cost, in dollars, of driving x kilometers?

(A) $\frac{xm}{1.6d}$ (B) $\frac{xd}{1.6m}$ (C) $\frac{1.6x}{md}$ (D) $\frac{1.6xd}{m}$

30.

SAT CHALLENGE · NO CALCULATOR NEEDED

Cube A and Cube B are made of the same material. The edge length of Cube A is 8 centimeters, and the mass of Cube B is 72.8% greater than the mass of Cube A. What is the edge length of Cube B, in centimeters?

(A) 10.5 (B) 13.8

(C) 9.6 (D) 9.9

31.

SAT CHALLENGE · CALCULATOR

A test vehicle accelerates at a constant 22 feet per second squared. Which of the following is equal to this acceleration in miles per minute squared? (1 mile = 5280 feet)

(A) 15 (B) 79,200

(C) $\frac{1}{4}$ (D) $\frac{1}{240}$

32.

SAT CHALLENGE · CALCULATOR

A crew spreads fertilizer on two kinds of turf. Type A turf receives 30 grams per square meter, and Type B turf receives 4 kilograms per 100 square meters. In all, the crew treats 70,000 square meters and uses 2.6 metric tons of fertilizer (1 metric ton = 1000 kilograms). How many square meters of Type A turf are treated?

33.

SAT CHALLENGE · CALCULATOR

A car's fuel economy is 40 miles per gallon. Use 1 mile = 1.6 kilometers and 1 gallon = 3.8 liters. How many liters of fuel does the car need to drive 304 kilometers?

34.

SAT CHALLENGE · CALCULATOR

A commuter train accelerates at a constant rate of 0.5 meters per second squared. What is this acceleration, in kilometers per hour squared? (1 kilometer = 1000 meters and 1 hour = 3600 seconds.)

35.

SAT CHALLENGE · NO CALCULATOR NEEDED

A table has 4 legs, each with the same ground-contact area of A square centimeters. With a box on it, the table presses on the floor with a pressure of 12 newtons per square centimeter on each leg. A 120-newton weight is then added to the box, and the table is tilted so that only 3 legs touch the floor. The pressure on each of those 3 legs is now 24 newtons per square centimeter. (Pressure is force divided by contact area, and the load is shared equally by the legs touching the floor.) What is the value of A ?

36.

SAT CHALLENGE · NO CALCULATOR NEEDED

A box contains only red, green and blue flash drives, 140 in all. The ratio of red drives to green drives is 7 to 4. The ratio of blue drives to drives that are not blue is 3 to 11. How many more red drives than blue drives are in the box?

37.

SAT CHALLENGE · CALCULATOR

A contractor paints the walls of a room with total wall area x square feet. Primer is applied to all of the wall area in one coat; one gallon of primer covers 400 square feet and costs \$20. Then paint is applied to $\frac{3}{5}$ of the wall area in two coats; one gallon of paint covers 300 square feet per coat and costs \$30. The contractor buys 10% more paint than the calculated amount, but buys exactly the calculated amount of primer. Gallons may be purchased in fractions. Which of the following functions gives the total cost $C(x)$, in dollars, of the primer and paint?

(A) $C(x) = 0.187x$

(B) $C(x) = 0.232x$

(C) $C(x) = 0.170x$

(D) $C(x) = 0.182x$

38.

SAT CHALLENGE · CALCULATOR

A site plan was drawn so that 1 centimeter on the plan represents 40 meters of ground. A copier then enlarged the plan to 125% of its size. On the enlarged copy, the parcel appears as a rectangle measuring 7.5 centimeters by 5 centimeters. What is the actual area of the parcel, in hectares? (1 hectare = 10,000 square meters.)

(A) 38.4

(B) 384

(C) 3.84

(D) 6

39.

SAT CHALLENGE · NO CALCULATOR NEEDED

A rectangular banner has width w and length $1.8w$, where $w > 0$. The banner is covered exactly, with no gaps or overlaps, by 4 identical square tiles. The side length of each tile is $k w \sqrt{5}$, where k is a positive constant. What is the value of k ?

40.

SAT CHALLENGE · CALCULATOR

A brass bushing is a solid right circular cylinder with diameter 60 millimeters and height 40 millimeters. A cylindrical hole of diameter 20 millimeters is drilled into one flat face, along the axis, to a depth of 25 millimeters. The density of brass is 7.8 grams per cubic centimeter. Which of the following is closest to the mass of the bushing, in grams? (1 cm = 10 mm)

(A) 784

(B) 821

(C) 882

(D) 82,090

41.

SAT CHALLENGE · CALCULATOR

A contractor finishes a 96-square-meter floor using two products. Primer P is applied in two coats, and each coat uses 125 milliliters per square meter. Sealant Q is applied in one coat and covers 5 square meters per liter. Every square meter receives exactly one of the two products, and the job used a total of 21.2 liters. How many square meters did the contractor cover with primer P?

42.

SAT CHALLENGE • CALCULATOR

A tram increases its speed at a constant rate from 54 kilometers per hour to 90 kilometers per hour in 20 seconds. What is the tram's acceleration, in meters per minute squared?

43.

SAT CHALLENGE • CALCULATOR

A car uses fuel at a rate of 6.0 liters per 100 kilometers on a highway and 10.0 liters per 100 kilometers in the city. On one trip the car travels 240 kilometers on the highway and 80 kilometers in the city. Using 1 mile = 1.6 kilometers and 1 gallon = 3.8 liters, which of the following is closest to the car's fuel efficiency for the whole trip, in miles per gallon?

- (A) 87 (B) 6
(C) 30 (D) 34

44.

SAT CHALLENGE · NO CALCULATOR NEEDED

On a map, 1 inch represents 1,500 feet. A rectangular parking lot appears on the map as 2.4 inches by 1.6 inches. Which of the following is the actual area of the lot, in square yards? (1 yard = 3 feet)

(A) 960,000

(B) 2,880,000

(C) 8,640,000

(D) 320,000

45.

SAT CHALLENGE · CALCULATOR

A cyclist rides a 90-kilometer route. She covers the first 60% of the distance at an average speed of 27 kilometers per hour, rests for 30 minutes, and then covers the rest of the route at an average speed of 24 kilometers per hour. What was her average speed for the whole trip, including the rest, in meters per second?

(A) 22.5

(B) 6.25

(C) 7.08

(D) 7.14

46.

SAT CHALLENGE · NO CALCULATOR NEEDED

A 84-gram sample of an alloy contains only copper, zinc and tin. The mass of the copper is 5 grams more than twice the mass of the zinc. The mass of the tin is half the combined mass of the copper and the zinc. What is the mass, in grams, of the tin in the sample?
